

Po-Chun Wu

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Education

Carnegie Mellon University

M.S. Electrical and Computer Engineering / Engineering and Technology Innovation Management

Pittsburgh, PA

May 2028

- **Ongoing Courses:** Fundamentals of Computer Systems, Optimization

National Taiwan University

B.S. Electrical Engineering; GPA: 4.16/4.3

Taipei, Taiwan

Jan 2026

- **Relevant Courses:** Discrete Mathematics, Data Structures, Algorithms, Machine Learning, SoC Verification

Projects

Whiskers Compiler and IDE – Personal Project

Jan 2026 – Feb 2026

- Built a Next.js-based compiler to convert a Scratch-inspired programming syntax into JavaScript.
- Designed a drag-and-drop interface to guide new learners in transitioning from Scratch to text-based programming.

LifeOS Automation System – NTU, Course Project (Team of 6)

Jan 2025 – Mar 2025

- Designed a backend Notion plugin to synchronize entries across different databases.
- Engineered a timestamp-based version-state model to resolve sync conflicts between Notion pages and MongoDB records.
- Developed a document-linking layer mapping a canonical schedule to multiple nodes for one-to-many synchronization.

AR Navigation Helmet – Best Maker Award, 2024 MakeNTU Competition (Team of 5)

May 2024

- Created a navigation assistant mounted inside a motorcycle helmet, using speech recognition and AR to help riders navigate.
- Designed a multi-threaded system in Python to handle asynchronous voice input, API requests, and OLED rendering.

NTU Pair – Personal Project

Dec 2022 – Feb 2023

- Built a full-stack dating app using *vanilla JS*, reaching 1,600+ verified users from NTU.
- Designed a matchmaking and recommendation algorithm to match users based on psychological traits.
- Re-architected the system with secure coding practices after mitigating XSS, DoS, and credential-stuffing attacks.

Work Experience

SLEKMED

Taipei, Taiwan (Remote)

Product Developer (Part-Time)

Aug 2023 – Mar 2026

- **Note-to-Video Generator:** Developed a web-based note-to-video generator using Marp, OpenAI API, ElevenLabs API, and FFmpeg, enabling medical experts to create high-quality educational videos from Markdown notes in less than 10 minutes.
- **Medical Simulation Platform:** Built a Next.js-based simulation module for medical diagnostic training in collaboration with Taipei Medical University peers, gaining 4,200+ followers and securing funding from the Taiwan Ministry of Education.
- **Virtual Patient Chatbot:** Implemented an LLM-based chatbot as a virtual patient, contributing to a 1st-place finish among 40+ teams in the 2024 CTCL AI Competition (12K USD prize).

Skills

- **Programming:** Python, JavaScript/TypeScript, C++
- **Frameworks/Tools:** React/Next.js, Node.js, Asyncio, MongoDB, Docker, Git
- **Specialties:** System design, algorithm design, full-stack development

Leadership

Founder and President, NTU Learning Optimization Club

Aug 2023 – Jun 2025

- Researched learning methodologies, delivered 8 nationwide speeches, and advised 200+ students on study habits.

Makerspace Administrator, NTUEE Student Association Academic Department

Sep 2023 – Dec 2023

- Organized 25 academic workshops and taught microcontroller programming to electrical engineering undergraduates.

Project Lead, NTUEE Virtual Makerspace

May 2023 – Aug 2023

- Led a 25-member team to build a React/Node.js inventory and reservation system for the university student makerspace.